

Forensic Disk Imaging using Python

Python Program:

```
import hashlib

def sha256_of(filename):
    with open(filename,"rb") as f:
        return hashlib.sha256(f.read()).hexdigest()

with open("original_disk.img","wb") as f:
    f.write(b"HEADER"+bytes(range(256))*4+b"FOOTER")

with open("original_disk.img","rb") as src:
    with open("forensic_copy.img","wb") as dst:
        dst.write(src.read())

original_hash=sha256_of("original_disk.img")
copy_hash=sha256_of("forensic_copy.img")

print("Original image hash:",original_hash)
print("Forensic copy hash :",copy_hash)
if original_hash==copy_hash:
    print("Match: VERIFIED - exact bit-stream copy")
else:
    print("MISMATCH - copy corrupted")
```

Sample Output:

```
Original image hash:
d50630bee2bf9926344781042d10cb09377730818efda8e80f9eeec21008a9b0
Forensic copy hash :
d50630bee2bf9926344781042d10cb09377730818efda8e80f9eeec21008a9b0
Match: VERIFIED - exact bit-stream copy
```

Explanation:

This program demonstrates a simple forensic disk imaging process. It creates a sample disk image, makes an exact bit-for-bit copy, and then calculates the SHA-256 hash of both the original image and the copied image. If both hash values are identical, it confirms that the forensic copy is an exact and unaltered duplicate of the original data. This technique is commonly used in digital forensics to preserve evidence integrity.